Sahaj Bhadja

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Summary

Application specialist with 2+ years analyzing requirements and building, testing, documenting, deploying, and supporting business systems using **Python**, Java, TypeScript, and SQL. Experienced in user-facing workflows, database models, troubleshooting, and application maintenance.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Software Engineering, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Analyzed user requirements and developed **React**, Python, and Azure SQL application workflows for 500+ users, implementing screen logic, role-based permissions, reports, and indexed queries that reduced latency to 350 milliseconds.
- Maintained application workflows processing 5,000+ monthly jobs, documenting functional behavior and resolving interruptions through reconciliation, structured logging, monitoring, recovery procedures, regression checks, and controlled CI/CD-backed implementation releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed and implemented **React** screens, forms, reports, file collection, and grading workflows for 150+ users, translating customer needs into testable functionality supporting 10,000+ daily requests through Python services.
- Supported users and evaluated application functionality through validation, progress notifications, defect recovery, and documented status guidance for 1,000+ files, reducing turnaround 60% while preserving data quality and usability.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Conducted cross-platform build, installation, regression, and integration testing across Windows, Linux, and macOS, maintaining 20+ automated suites, CSV reports, coverage summaries, and failure logs for structured application reviews.
- Troubleshot and enhanced object-oriented applications using Python, Java, SQL, REST APIs, caching, and pagination, improving response times 30% and processing efficiency 68% while reducing recurring operational issues.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js applications across eight business modules, translating requirements into screen layouts, reports, resource, billing, notification, and workflow functionality spanning more than two million records.
- Modeled application data in Azure SQL and coordinated unit, integration, and end-to-end testing with Jest, Selenium, Jenkins, and Sentry, cutting release time 50% while strengthening production support.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Conferred with customers to review business needs and documented functional requirements for authentication, booking, payment, notifications, and reporting, mapping 12+ integration points before presenting the technical design to stakeholders.
- Built and supported React, TypeScript, Flask, and Firebase applications handling 50,000+ records, troubleshooting status, payment, and synchronization issues while maintaining sub-200-millisecond interactions across mobile and web systems.

Skills

Applications: Python, Java, TypeScript, JavaScript, React.js, Node.js, Flask, HTML, CSS, screen layouts, forms, reports, REST APIs, JSON
Databases: SQL, Azure SQL, SQL Server, PostgreSQL, MySQL, MongoDB, relational modeling, schema design, indexing, query optimization, data validation
Analysis & Design: Business needs analysis, user requirements, functional specifications, technical design, program logic, prototyping, workflow design, implementation planning
Testing & Support: UAT support, unit testing, integration testing, regression testing, performance testing, debugging, troubleshooting, recovery procedures, production support, user guidance
Delivery: Git, Jenkins, GitHub Actions, CI/CD, Docker, Agile/Scrum, structured reviews, technical documentation, process documentation, security and access controls

Projects

HMIS | Live | *React, TypeScript, Role-Based Access, Sentry*
- Built a role-based management console with scheduling, controlled workflows, dashboards, notifications, validation, and Sentry monitoring, reducing administrative workload 30% while supporting secure access and operational visibility for users.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker*
- Built a database workspace supporting 10+ SQL platforms with schema exploration, query generation, validation, and export workflows, helping users transform structured data requests into reliable and reusable database operations.

FinNexus | Live | *Next.js, TypeScript, Dashboards, WebSockets*
- Created an operational finance dashboard for 50,000+ users and one million transactions, integrating secure access, real-time updates, advanced filtering, notifications, and reporting views across sensitive data workflows.